

Slime Mold Simulation: Theory & Data Analysis

1 Theoretical Derivation of Classical Keller-Segel

1.1 Diffusion Equation

We start from the continuity equation,

$$\frac{\partial}{\partial t} \int_V C dV = - \int_S \vec{j} \cdot d\vec{\sigma} + \int_V f dV \quad (1)$$

where C is the concentration, $\vec{j}$ is the flux, and f is the source term. Using the divergence theorem, we get the conservation equation,

$$\int_V \left[\frac{\partial C}{\partial t} + \nabla \cdot \vec{j} - f \right] dV = 0 \implies \frac{\partial C}{\partial t} + \nabla \cdot \vec{j} = f. \quad (2)$$

Using Fick's law of diffusion, the flux is proportional to the gradient of the field: $\vec{j} = -D\nabla C$. Thus, we get

$$\frac{\partial C}{\partial t} = f + \nabla \cdot (D\nabla C). \quad (3)$$

In our specific model for chemical concentration c , we get

$$\frac{\partial c}{\partial t} = D_{\text{chem}} \nabla^2 c - \gamma c + \alpha \rho \quad (4)$$

with diffusion constant D_{chem} , degradation rate γ and secretion rate α proportional to cell density ρ .

1.2 Chemotaxis

We use a continuous model for cell concentration here. Let's suppose the presence of a chemical concentration gives rise to the movement of cells. The flux of cells is given by:

$$\vec{J} = \vec{J}_{\text{diff}} + \vec{J}_{\text{chemo}} \quad (5)$$

where $\vec{J}_{\text{diff}} = -D_{\text{cell}} \nabla \rho$ and the chemotactic flux is $\vec{J}_{\text{chemo}} = \rho \chi(c) \nabla c$. Here, $\chi(c)$ is the chemotactic sensitivity function.

The cell population also follows the continuity equation

$$\frac{\partial \rho}{\partial t} + \nabla \cdot \vec{J} = f(\rho). \quad (6)$$

Substituting the flux terms gives

$$\frac{\partial \rho}{\partial t} = f(\rho) + \nabla \cdot (D_{\text{cell}} \nabla \rho) - \nabla \cdot (\rho \chi(c) \nabla c). \quad (7)$$

For constant parameters and no cell growth or production ($f(\rho) = 0$), the equation is simplified to

$$\frac{\partial \rho}{\partial t} = D_{\text{cell}} \nabla^2 \rho - \chi \nabla \cdot (\rho \nabla c). \quad (8)$$

In a more realistic model, chemotactic sensitivity function is usually modelled as $\chi(\rho, c) = \frac{b\rho}{(a+c)^2}$, where a and b are parameters to be determined. For simplification, χ is set to be constant in this simulation.

Summary of Equations:

$$1. \quad \frac{\partial c}{\partial t} = D_{\text{chem}} \nabla^2 c - \gamma c + \alpha \rho \quad (9)$$

$$2. \quad \frac{\partial \rho}{\partial t} = D_{\text{cell}} \nabla^2 \rho - \chi \nabla \cdot (\rho \nabla c) \quad (10)$$

Generally, diffusion is a stabilizing force while chemotaxis is destabilizing. The balance between stabilization and destabilization can result in steady states or unsteady, wave-like patterns. If chemotaxis is too strong, the solution blows up.

1.3 Linear Stability Analysis for the Keller-Segel Model

To find the collapse threshold, we check if the uniform distribution is stable against tiny fluctuations:

$$\rho(r, t) = \rho_0 + \delta \rho \quad \text{and} \quad c(r, t) = c_0 + \delta c \quad (11)$$

Plugging these into Equation 8 and ignoring the second-order term $\delta \rho \nabla \delta c$:

$$\frac{\partial \delta \rho}{\partial t} = D_{\text{cell}} \nabla^2 \delta \rho - \chi \rho_0 \nabla^2 \delta c \quad (12)$$

For the chemical field (Equation 4), we assume a quasi-steady state that the chemical diffuses much faster than the cells ($\frac{\partial c}{\partial t} \approx 0$). Thus we get

$$0 = D_{\text{chem}} \nabla^2 \delta c - \gamma \delta c + \alpha \delta \rho. \quad (13)$$

By doing this, we linearized the two PDEs! Now we can solve Equation 12 and 13 using usual method (Fourier transform for spatial domain and Laplace transform for the time domain which is initial value problem).

Rules:

- FT: $\mathcal{F}(\nabla^2 f) = -k^2 \tilde{f}$
- LT: $\mathcal{L}\left(\frac{df}{dt}\right) = \omega \tilde{f} - f(t=0)$

Transforming two equations gives

$$\omega \delta \tilde{\rho} - \delta \tilde{\rho}_i = -k^2 (D_{\text{cell}} \delta \tilde{\rho} - \chi \rho_0 \delta \tilde{c}) \quad (14)$$

$$0 = -D_{\text{chem}} k^2 \delta \tilde{c} - \gamma \delta \tilde{c} + \alpha \delta \tilde{\rho} \quad (15)$$

where $\delta \rho_i = \delta \rho(t=0)$. We can solve them as 2×2 matrix problem

$$\begin{pmatrix} \omega + D_{\text{cell}} k^2 & -\chi \rho_0 k^2 \\ -\alpha & \gamma + D_{\text{chem}} k^2 \end{pmatrix} \begin{pmatrix} \delta \tilde{\rho} \\ \delta \tilde{c} \end{pmatrix} = \begin{pmatrix} \delta \tilde{\rho}_i \\ 0 \end{pmatrix}. \quad (16)$$

To solve for $\delta \tilde{\rho}$, invert the matrix and we get

$$\delta \tilde{\rho}(k, \omega) = \frac{\delta \tilde{\rho}_i}{\omega + D_{\text{cell}} k^2 - \frac{\chi \rho_0 \alpha k^2}{\gamma + D_{\text{chem}} k^2}}. \quad (17)$$

Using the inverse Laplace transform $\mathcal{L}^{-1}\left(\frac{1}{s-a}\right) = e^{at}$, the time-domain solution is

$$\delta \tilde{\rho}(k, t) = \delta \tilde{\rho}_i(k) e^{\omega(k)t} \quad (18)$$

where the dispersion relation is obtained as

$$\omega(k) = -D_{\text{cell}} k^2 + \frac{\chi \rho_0 \alpha k^2}{\gamma + D_{\text{chem}} k^2}. \quad (19)$$

1.4 Dispersion Relation and Critical Density

- When $\omega < 0 \implies$ exponential decay (stable mode).
- When $\omega > 0 \implies$ exponential growth (unstable mode $\implies$ collapse condition).

Setting $\omega(k) > 0$:

$$\left(\frac{\chi \rho_0 \alpha}{\gamma + D_{\text{chem}} k^2} - D_{\text{cell}} \right) k^2 > 0, \quad (20)$$

assuming $k \neq 0$, the minimum ρ_0 for instability emerges when $k \rightarrow 0$, so we get

$$\rho_0 > \frac{D_{\text{cell}}(\gamma + D_{\text{chem}} k^2)}{\chi \alpha} \implies \rho_c = \frac{D_{\text{cell}} \gamma}{\chi \alpha} \quad (21)$$

which is the critical density for collapse.

1.5 Find the Cluster Separation

To find the cluster separation, we need the fastest growing k -mode that overtakes the other modes. We find this by maximizing the dispersion relation $\frac{d\omega}{dk^2} = 0$:

$$-D_{\text{cell}} + \frac{\chi \rho_0 \alpha (\gamma + D_{\text{chem}} k^2) - \chi \rho_0 \alpha k^2 D_{\text{chem}}}{(\gamma + D_{\text{chem}} k^2)^2} = 0 \quad (22)$$

This simplifies to:

$$D_{\text{cell}}(\gamma + D_{\text{chem}}k^2)^2 = \chi\rho_0\alpha\gamma \implies D_{\text{chem}}k^2 + \gamma = \sqrt{\frac{\chi\rho_0\alpha\gamma}{D_{\text{cell}}}} \quad (23)$$

Solving for k_{max}^2 :

$$k_{\text{max}}^2 = \frac{1}{D_{\text{chem}}} \left(\sqrt{\frac{\chi\rho_0\alpha\gamma}{D_{\text{cell}}}} - \gamma \right) \quad (24)$$

Using the critical density $\rho_c = \frac{D_{\text{cell}}\gamma}{\chi\alpha}$, the maximum wavelength is

$$\lambda_{\text{max}} = 2\pi\sqrt{\frac{D_{\text{chem}}}{\gamma}} \left(\sqrt{\frac{\rho_0}{\rho_c}} - 1 \right)^{-1/2} = 2\pi\lambda \left(\sqrt{\frac{\rho_0}{\rho_c}} - 1 \right)^{-1/2} \quad (25)$$

where $\lambda = \sqrt{\frac{D_{\text{chem}}}{\gamma}}$ is the screening length.

1.6 Steady-state Density Profile

Starting from the general equation at steady state ($\frac{\partial\rho}{\partial t} = 0$):

$$\nabla \cdot (D_{\text{cell}}\nabla\rho - \chi\rho\nabla c) = 0 \implies D_{\text{cell}}\nabla\rho - \chi\rho\nabla c = 0 \quad (26)$$

Integrating spatially gives

$$\frac{D_{\text{cell}}}{\chi} \frac{\nabla\rho}{\rho} = \nabla c \implies \nabla \left(\frac{D_{\text{cell}}}{\chi} \ln\rho - c \right) = 0 \quad (27)$$

$$\frac{D_{\text{cell}}}{\chi} \ln\rho = c + A \implies \rho(r) = \rho_{bg} \exp\left(\frac{\chi c(r)}{D_{\text{cell}}}\right). \quad (28)$$

This is exactly the Boltzmann distribution where $U(r) = -\chi c(r)$ and $k_B T = D_{\text{cell}}$. This equation doesn't give the actual distribution, but shows the relation between $\rho(r)$ and $c(r)$.

2 Data Analysis

2.1 Phases of the Evolution of the System

- **Phase 1: Linear phase.** This is the phase where we can use linear perturbation theory. When the simulation first starts, the agents are roughly uniform and k_{max} dictates the system.
- **Phase 2: Non-linear phase.** The linear approximation breaks down completely. Each cluster behaves like a gravitational well acting with long-range forces.

2.2 Finding the Separation Between Clusters

The best way to do this is working in Fourier space. Essentially, I find the power spectrum of the cell distribution to find the peak mode. The problem is: which time step am I supposed to look at to pick the maximum k -mode? I need to determine when the linear regime occurs. The best way is to use $\rho_{\text{max}} < 1.1\rho_0$ as a threshold for the linear regime.

2.3 Issue with the Initial Setup

The number of cells per pixel follows a binomial distribution:

$$P(k) = \binom{N}{k} p^k (1-p)^{N-k} \quad \text{where } p = \frac{1}{\text{total pixels}} \quad (29)$$

In the limit of large N , this becomes a Poisson distribution with mean $\lambda = N \times p = \rho_0$, simply meaning our grids are small enough to treat them as smooth continuous space where the variance is $\sigma^2 = \rho_0$.

The percentage fluctuation is

$$\frac{\delta\rho}{\rho_0} = \frac{\sigma}{\rho_0} = \frac{1}{\sqrt{\rho_0}} \quad (30)$$

Thus if we set ρ_0 too small (e.g., 10 per pixel), the noise is too large to ever be in the linear regime. We want $\frac{\delta\rho}{\rho_0} \approx 0.03 \implies \rho_0 = 1000$.

2.4 Calculating Physical Length

The discrete Fourier transform is given by:

$$\mathcal{F}(k_x, k_y) = \sum_{x=0}^{N_x-1} \sum_{y=0}^{N_y-1} \delta\rho(x, y) e^{-i2\pi(k_x \frac{x}{N_x} + k_y \frac{y}{N_y})} \quad (31)$$

In continuous space, $\vec{k}' = 2\pi \left(\frac{1}{\lambda_x} \hat{i} + \frac{1}{\lambda_y} \hat{j} \right)$. Thus we have

$$2\pi \sqrt{\frac{k_x^2}{N_x^2} + \frac{k_y^2}{N_y^2}} = |\vec{k}'| = \frac{2\pi}{\lambda} \quad (32)$$

If we assume a square grid $N_x = N_y = N$, then

$$\frac{1}{N} \sqrt{k_x^2 + k_y^2} = \frac{k}{N} = \frac{1}{\lambda} \implies \lambda_{\max} = \frac{N}{k} \quad \text{and} \quad \delta\lambda_{\max} = \lambda_{\max} \frac{\delta k_{\max}}{k_{\max}}. \quad (33)$$

From the context of the simulation parameters: $D_{\text{chem}} = 1$, $\gamma = 0.1$, $\alpha = 1$, $\chi = 2$, $D_{\text{cell}} = D_r = 0.5$.

$$\lambda_{\max} = 2\pi \sqrt{\frac{1}{0.1}} \left(\sqrt{10000 \times \frac{2 \times 1}{0.5 \times 0.1}} - 1 \right)^{-1/2} = 0.79 \quad (34)$$

There is a problem here. $\lambda_{\max} = 0.79$ but $dx = 1$, which means we can't resolve the wavelength. Theoretically, we must have at least $\lambda = 2$ (two pixels) to see a peak and trough. It's safe to explore from $\lambda \sim 3$. Solutions:

- Keep biological parameters and increase resolution of the simulation.
- Keep the resolution but move biological parameters.

2.5 Using MLE to Extract $k_{\max}$

To use Maximum Likelihood Estimation (MLE), we need to know the likelihood function for a single data point.

The complex coefficient from the Fourier transform is

$$Z(\vec{k}) = \int \delta\rho(\vec{r}) e^{i\vec{k}\cdot\vec{r}} d\vec{r} = X + iY. \quad (35)$$

By the Central Limit Theorem, we can assume Z is randomly distributed that

$$X, Y \sim \mathcal{N}(0, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{X^2 \text{ or } Y^2}{2\sigma^2}\right). \quad (36)$$

Therefore, the joint probability of getting X and Y is

$$f(X, Y) dX dY = \frac{1}{2\pi\sigma^2} \exp\left(-\frac{X^2 + Y^2}{2\sigma^2}\right) dX dY \quad (37)$$

By doing a coordinate transformation $(X, Y) \rightarrow (P, \theta)$ where $P = X^2 + Y^2$ and $dX dY = \frac{1}{2} dP d\theta$. Integrating over the phase θ gives

$$f(P) dP = \frac{1}{2\sigma^2} \exp\left(-\frac{P}{2\sigma^2}\right) dP \quad (38)$$

which is exponential distribution.

But what is σ^2 ? From previous relation, $\sigma^2 = \langle X^2 \rangle = \langle Y^2 \rangle$, thus we can say

$$\sigma^2 = \frac{1}{2} \langle X^2 + Y^2 \rangle = \frac{1}{2} S(k) \quad (39)$$

where $S(k) = \langle P(k) \rangle$, the expected power for given k . Here We model $S(k)$ as a Gaussian curve

$$S(k) = A \exp\left(-\frac{(k - k_{\max})^2}{2w^2}\right) + C \quad (40)$$

where A, k, w, C are adjustable parameters. So our likelihood function can be written as

$$f(P|S(k)) = \frac{1}{S} \exp\left(-\frac{P}{S}\right). \quad (41)$$

Then log-likelihood function of getting a set of data points becomes

$$\ln \mathcal{L} \propto \sum_i^N \left[-\ln S(k_i) - \frac{P_i}{S(k_i)} \right]. \quad (42)$$

We maximize this function to find $k_{\max}$.

Important facts:

1. For a given k , the most probable data point is actually $P = 0$. 2. Trying to fit the Gaussian curve by binning the data points with symmetric Gaussian error can lead to significant flaw as the distribution of the data points are highly skewed.

Extra derivation: Why can we assume $\sigma_{X^2} = \sigma_{Y^2}$

In short, it's because the system is translationally symmetric.

$$X = \int \delta\rho(\vec{r}) \cos(\vec{k} \cdot \vec{r}) d\vec{r}, \quad Y = \int \delta\rho(\vec{r}) \sin(\vec{k} \cdot \vec{r}) d\vec{r} \quad (43)$$

then

$$\sigma_X^2 = \langle X^2 \rangle = \iint \delta\rho(\vec{r}) \delta\rho(\vec{r}') \cos(\vec{k} \cdot \vec{r}) \cos(\vec{k} \cdot \vec{r}') d\vec{r} d\vec{r}' \quad (44)$$

$$= \frac{1}{2} \iint \delta\rho(\vec{r}) \delta\rho(\vec{r}') \left[\cos(\vec{k} \cdot (\vec{r} - \vec{r}')) + \cos(\vec{k} \cdot (\vec{r} + \vec{r}')) \right] d\vec{r} d\vec{r}' \quad (45)$$

$$= \frac{1}{2} \iint \delta\rho(\vec{r}) \delta\rho(\vec{r}') \cos(\vec{k} \cdot (\vec{r} - \vec{r}')) d\vec{r} d\vec{r}' \quad (46)$$

as our system must be independent on the absolute coordinates $\vec{r} + \vec{r}'$ and only dependent on the relative coordinates $\vec{r} - \vec{r}'$. Similarly

$$\sigma_Y^2 = \langle Y^2 \rangle = \iint \delta\rho(\vec{r}) \delta\rho(\vec{r}') \sin(\vec{k} \cdot \vec{r}) \sin(\vec{k} \cdot \vec{r}') d\vec{r} d\vec{r}' \quad (47)$$

$$= \frac{1}{2} \iint \delta\rho(\vec{r}) \delta\rho(\vec{r}') \left[\cos(\vec{k} \cdot (\vec{r} - \vec{r}')) - \cos(\vec{k} \cdot (\vec{r} + \vec{r}')) \right] d\vec{r} d\vec{r}' \quad (48)$$

$$= \frac{1}{2} \iint \delta\rho(\vec{r}) \delta\rho(\vec{r}') \cos(\vec{k} \cdot (\vec{r} - \vec{r}')) d\vec{r} d\vec{r}' \quad (49)$$

Thus $\sigma_X^2 = \sigma_Y^2$